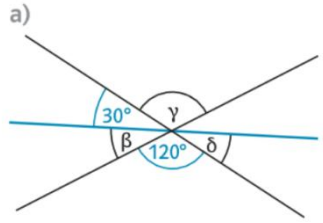
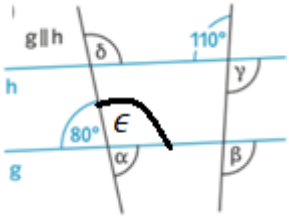
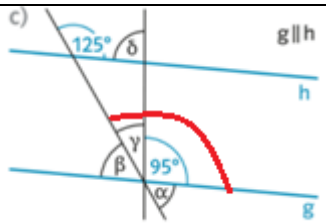
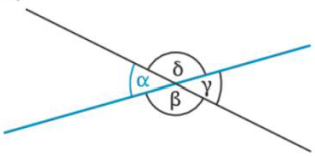
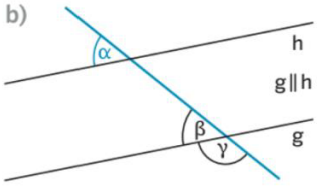
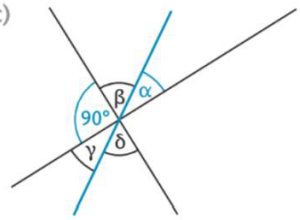
Thema 1 Lösungen

1. Seite 114 Nr. 6

	$\beta = 30°$ $\gamma = 120°$ $\delta = 30°$	$(weil\ es\ ein\ Nebenwinkel\ von$ $(120° + 30° = 150°))\ ist$ $\rightarrow\ Damit\ gilt\ für\ \beta = 180° - 150° = 30°$ $Scheitelwinkel\ von\ 120°$ $Scheitelwinkel\ von\ 30°$
	$\alpha = 80°$ $\beta = 110°$ $\gamma = 110°$ $\delta = 100°$ $\epsilon = 100°$	$Scheitelwinkel\ von\ 80°$ $Stufenwinkel\ von\ \gamma$ $Scheitelwinkel\ von\ 110°$ $Scheitelwinkel\ von\ \boldsymbol{\epsilon}$ Hilfswinkel ϵ mit Nebenwinkel von 80°
	$\alpha = 55°$ $\rightarrow \beta = 55°$ $\gamma = 30°$ $\delta = 85°$	$\alpha\ Nebenwinkel\ von\ (\boldsymbol{\gamma + 95}) = \boldsymbol{125°}$ $\rightarrow \alpha = 180° - 125° = 55°$ $\beta\ Scheitelwinkel\ von\ \alpha$ $\boldsymbol{\gamma + 95°}\ \ als\ Stufenwinkel\ von\ 125°$ $\rightarrow \boldsymbol{\gamma + 95°} = 125°\ \ \vert - 95°$ $<--->\ \ \gamma = 30°$ $\delta\ Stufenwinkel\ von\ (\beta + \gamma)$ $\rightarrow \delta = (\beta + \gamma) = 55° + 30° = 85°$

2. S. 144 Nr. 7

7 Gib die Winkel β, γ und δ mithi a) 	$\beta = 180° - \alpha$ $\gamma = \alpha$ $\delta = 180° - \alpha$
b) 	$\beta = \alpha$ $\gamma = 180° - \beta = 180° - \alpha$
c) 	$\beta = 90° - \alpha$ weil $\beta + 90° + \alpha = 180°$ $\gamma = \alpha$ Scheitelwinkel von α $\delta = \beta = 90° - \alpha$ Scheitelwinkel von β

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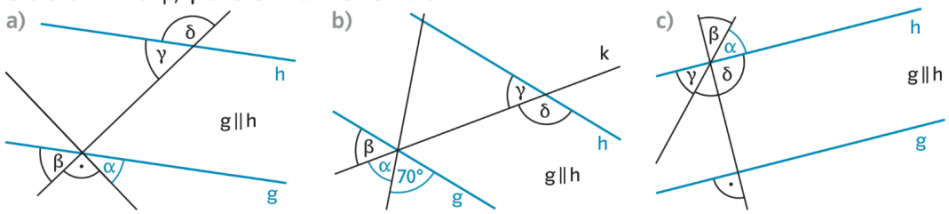
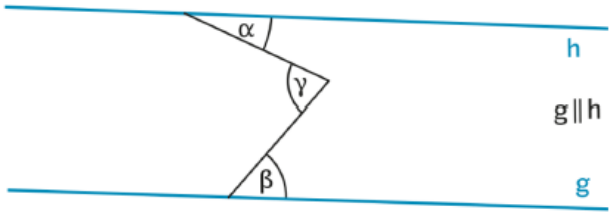
8 Gib die Winkel β, γ und δ mithilfe von α an.

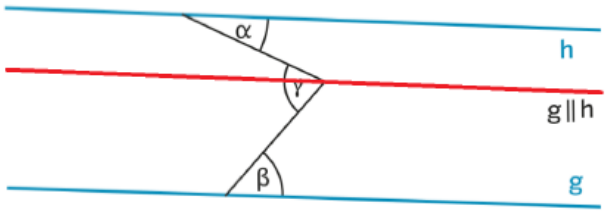


$\beta = 90° - \alpha$ $\gamma = 90° - \alpha$ $\delta = 180° - \gamma$ $= 180° - (90° - \alpha)$ $= 90° + \alpha$	$\beta = 180° - (70° + \alpha)$ $= 110° - \alpha$ $\gamma = \beta = 110° - \alpha$ $\delta = \alpha + 70°$	$\beta = 90° - \alpha$ weil Scheitelwinkel zu rechtem Winkel auch 90° → $\beta + \alpha = 90°$ Stufenwinkel $\gamma = \alpha$ Scheitelwinkel $\delta = 90°$

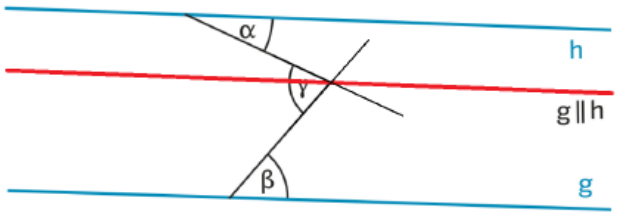
3. Seite 115 Nr. 9



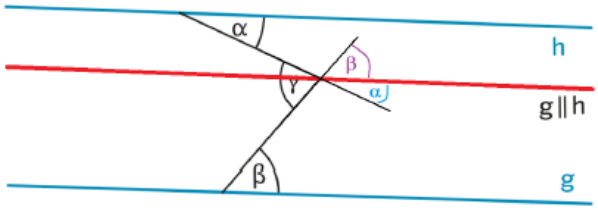
Parallele zu h und g einzeichnen



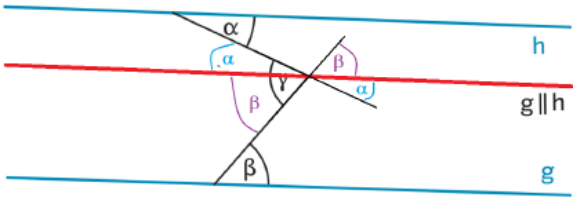
Strecken verlängern



Stufenwinkel einzeichnen (α *und* β)



Scheitelwinkel von den neuen Winkeln einzeichnen



$$\gamma = \alpha + \beta$$

Gamma ist also die Summe von α und β